

CS492C: Diffusion and Flow Models

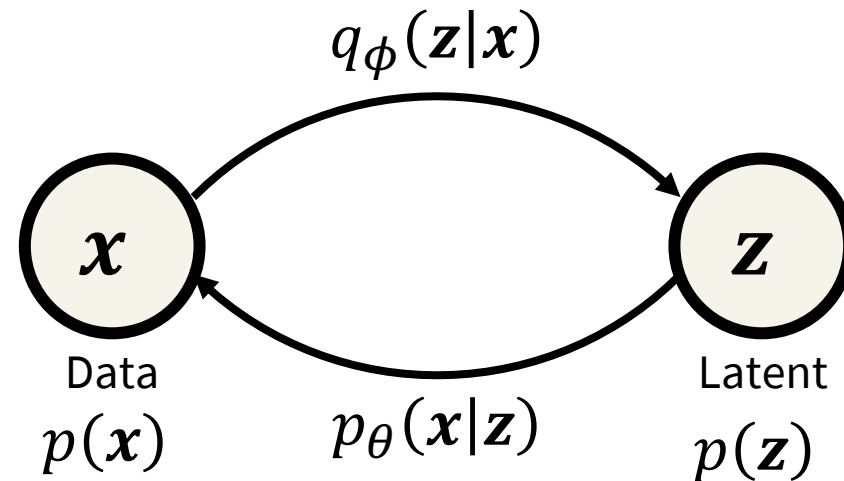
Denoising Diffusion Probabilistic Models 2

LECTURE 4
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Fall 2025
KAIST

Previously in CS492C

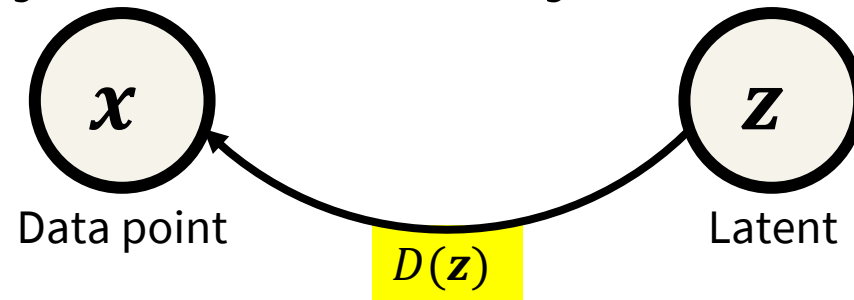
$$\overset{\text{Posterior Encoder}}{p(\mathbf{z}|\mathbf{x})} = \frac{\overset{\text{Likelihood Decoder}}{p(\mathbf{x}|\mathbf{z})} \overset{\text{Prior Latent}}{p(\mathbf{z})}}{\underset{\text{Marginal Data}}{p(\mathbf{x})}}$$



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In **VAE**, the likelihood distribution $p_{\theta}(\mathbf{x}|\mathbf{z})$ is modeled as a simple Gaussian distribution $\mathcal{N}(\mathbf{x}; D_{\theta}(\mathbf{z}), \sigma^2 I)$, which may limit expressiveness and fail to fully capture the data distribution:

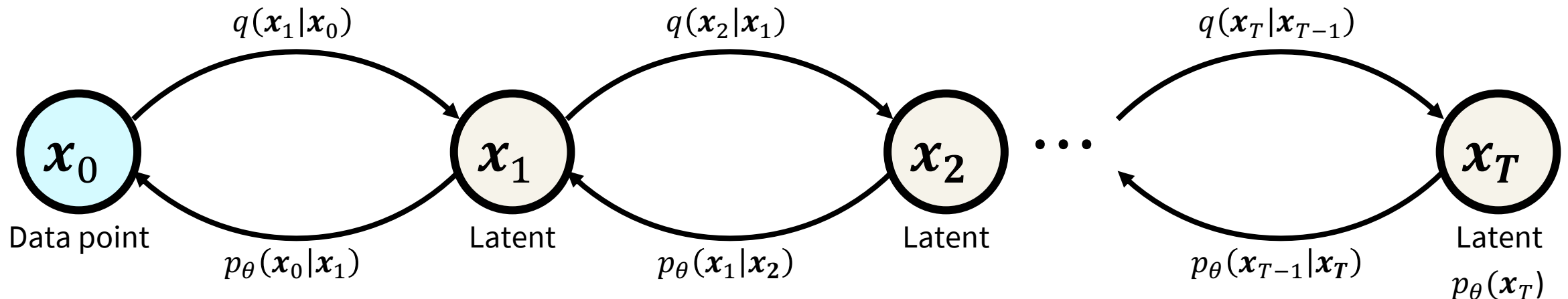
$$p(\mathbf{x}) = \int p(\mathbf{x}, \mathbf{z}) d\mathbf{z} = \int p_{\theta}(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) d\mathbf{z}$$



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Variational Diffusion Models:

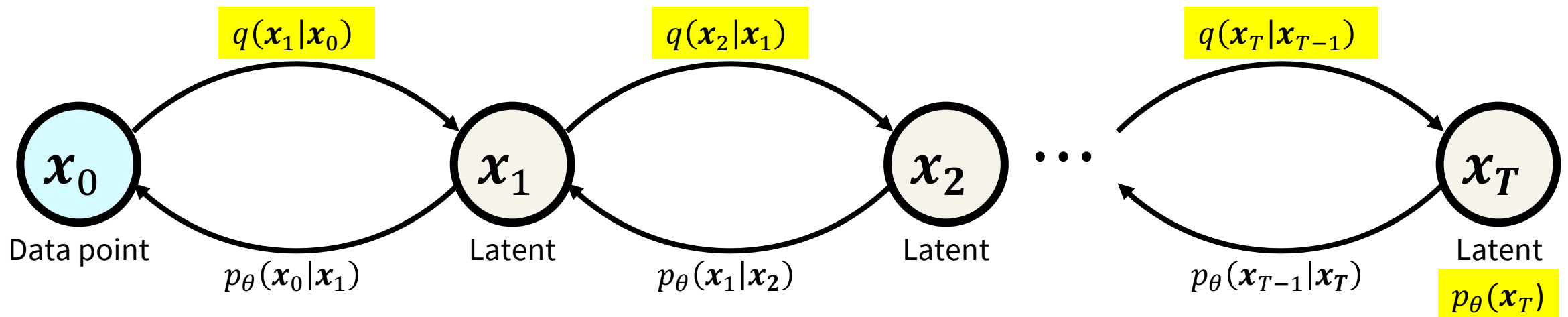
- Both forward (encoding) and reverse (decoding) processes are **sequential** processes (with a sequence of latent variables).



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Variational Diffusion Models:

1. Both forward (encoding) and reverse (decoding) processes are **sequential** processes (with a sequence of latent variables).
2. The **forward process** is not learned but **predefined**.



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Variational Diffusion Models:

1. Both forward (encoding) and reverse (decoding) processes are **sequential** processes (with a sequence of latent variables).
2. The **forward process** is not learned but predefined.
3. The **dimensions** of the latent variables and the input data are the **same**.

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Denoising Diffusion Probabilistic Models (DDPM):

1. The sequential forward and reverse processes are **Markovian** processes:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0)$$

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Denoising Diffusion Probabilistic Models (DDPM):

1. The sequential forward and reverse processes are **Markovian** processes:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0)$$

2. The **forward transition distribution** is defined as follows:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$$

where $\{\beta_t \in (0, 1)\}_{t=1}^T$ and $\beta_1 \leq \beta_2 \leq \dots \leq \beta_T$.

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Decompose the negative ELBO in a **different** way :

$$-\log p(\mathbf{x}_0) = -\log \int p_\theta(\mathbf{x}_{0:T}) d\mathbf{x}_{1:T} \leq \mathbb{E}_{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)} \right]$$

$$= \dots =$$

$$-\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)} [\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]$$

Reconstruction
term $\mathcal{L}_0$

$$+ D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T))$$

New prior matching
term $\mathcal{L}_T$

$$+ \sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

Denoising
matching
term $\mathcal{L}_{t-1}$

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- Reconstruction term $\mathcal{L}_0 = -\mathbb{E}_{q(\mathbf{x}_1|\mathbf{x}_0)}[\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)]:$
The same as with VAEs; **it is also negligible.**

- Prior matching term $\mathcal{L}_T = D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0) \parallel p(\mathbf{x}_T)):$
Converges to zero when $T \rightarrow \infty$.

- **Denoising matching term $\mathcal{L}_{t-1}$**

$$\mathcal{L}_{t-1} = \sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

Denoising Matching Term $\mathcal{L}_{t-1}$

Denoising Matching Term $\mathcal{L}_{t-1}$

$$\begin{aligned} & \sum_{t=2}^T \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))] \\ &= \mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))] \end{aligned}$$

The **variational** distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$ should be close to $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$ for each t .

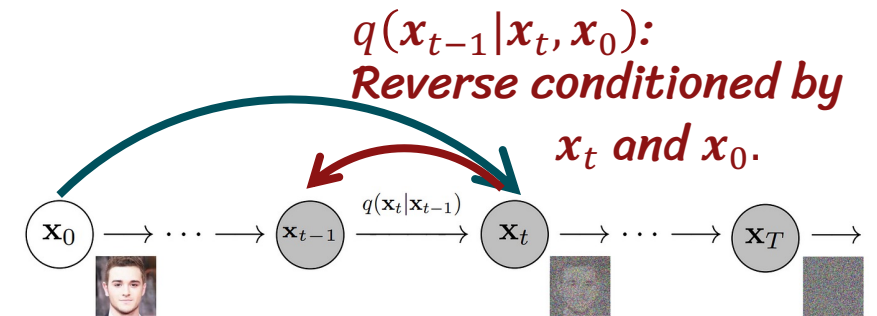
Denoising Matching Term $\mathcal{L}_{t-1}$

What is $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0)$?

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)}$$

Same as $q(\mathbf{x}_t|\mathbf{x}_{t-1})$, the forward transition, since it's a Markovian process.

We have seen how to compute these.



Quiz

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = q(\mathbf{x}_t|\mathbf{x}_{t-1}) \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)}$$

Q. Given that $q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}\mathbf{x}_{t-1}, (1 - \alpha_t)\mathbf{I})$ and $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$, what are the followings?

- $q(\mathbf{x}_t|\mathbf{x}_{t-1})$
- $q(\mathbf{x}_{t-1}|\mathbf{x}_0)$
- $q(\mathbf{x}_t|\mathbf{x}_0)$

Answer

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = q(\mathbf{x}_t|\mathbf{x}_{t-1}) \frac{q(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q(\mathbf{x}_t|\mathbf{x}_0)}$$

- $q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}\mathbf{x}_{t-1}, (1 - \alpha_t)\mathbf{I})$
- $q(\mathbf{x}_{t-1}|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_{t-1}}\mathbf{x}_0, (1 - \bar{\alpha}_{t-1})\mathbf{I})$
- $q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I})$

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$$

$$\begin{aligned} q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) &= q(\mathbf{x}_t | \mathbf{x}_{t-1}) \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_0)}{q(\mathbf{x}_t | \mathbf{x}_0)} \\ &\propto \exp \left(-\frac{1}{2} \left(\frac{(\mathbf{x}_t - \sqrt{\alpha_t} \mathbf{x}_{t-1})}{1 - \alpha_t} + \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_0)}{1 - \bar{\alpha}_{t-1}} - \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0)}{1 - \bar{\alpha}_t} \right) \right) \\ &= \dots \\ &= \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I}) \quad \text{Another normal distribution!} \end{aligned}$$

$$\text{where } \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 \text{ and } \tilde{\sigma}_t^2 = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t.$$

Forward Process Distributions

1. $q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I})$

where $\alpha_t = 1 - \beta_t$.

2. $q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$

where $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$.

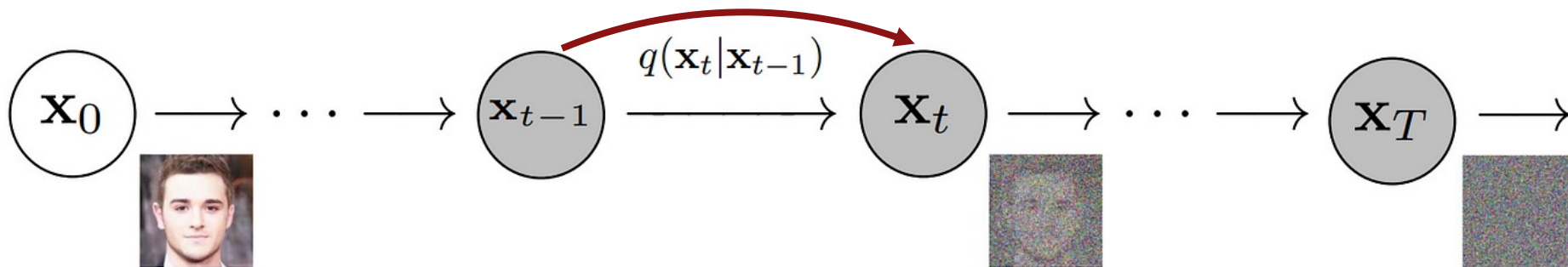
3. $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$

where $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0$ and $\tilde{\sigma}_t^2 = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$.

Forward Process Distributions

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I})$$

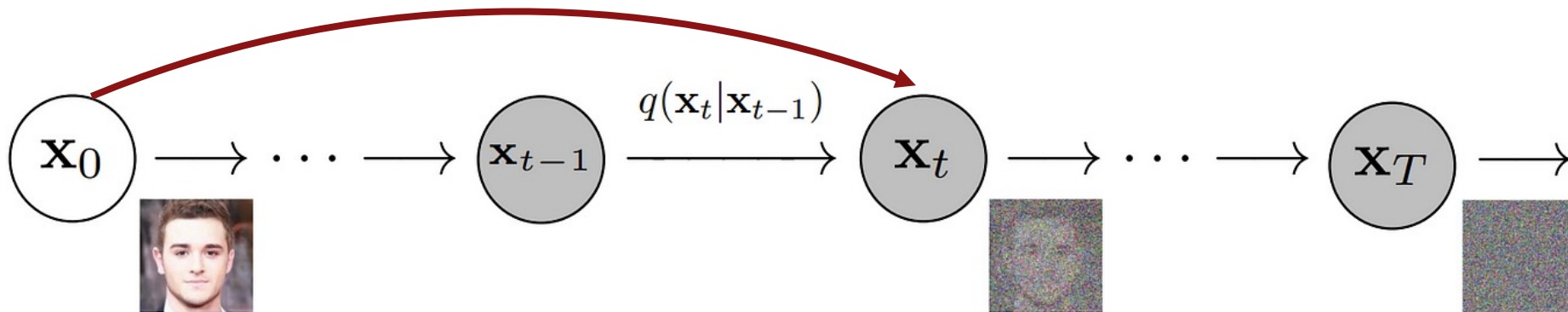
where $\alpha_t = 1 - \beta_t$.



Forward Process Distributions

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

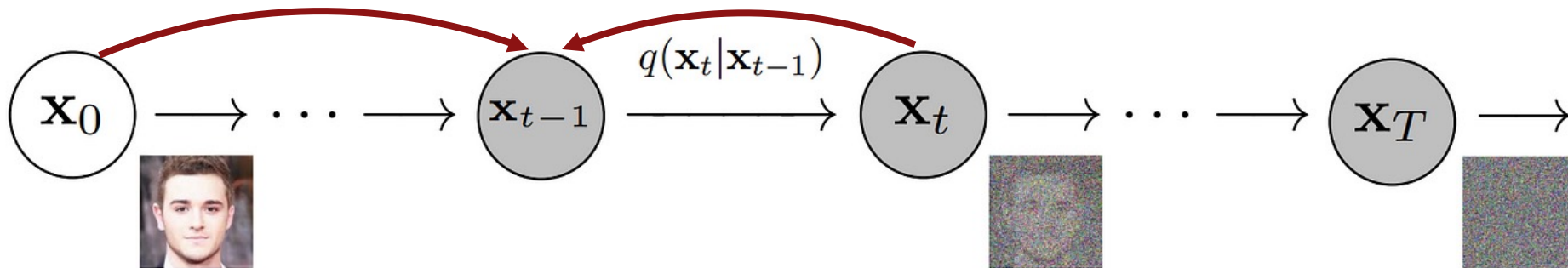
Where $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$.



Forward Process Distributions

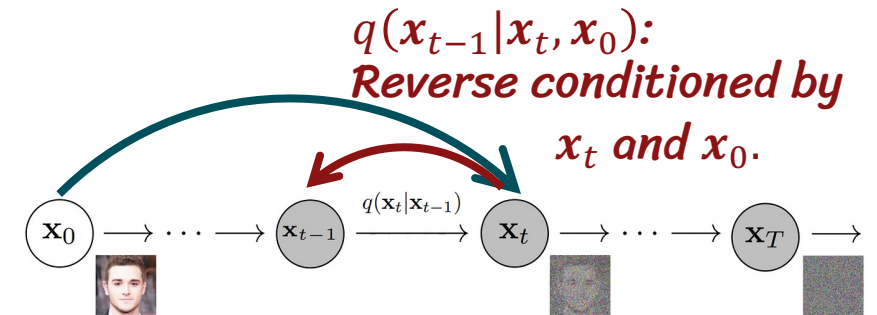
$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$$

where $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t} \mathbf{x}_0$ and $\tilde{\sigma}_t^2 = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t} \beta_t$.



$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$$

- The **mean** $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t} \mathbf{x}_0$ is a function of both $\mathbf{x}_t$ and $\mathbf{x}_0$.
- The **covariance** $\tilde{\sigma}_t^2 \mathbf{I} = \left(\frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t} \beta_t \right) \mathbf{I}$ is **predefined** from the user-defined $\{\beta_t\}_{t=1}^T$.



Denoising Matching Term $\mathcal{L}_{t-1}$

Back to the denoising matching term...

$$\mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

How to model the **variational** distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$?

Mean Predictor

- For $q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$,
the variance $\tilde{\sigma}_t^2$ is **given**.
- Hence, define the variational distribution $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ as

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu_\theta(\mathbf{x}_t, t), \tilde{\sigma}_t^2 \mathbf{I}),$$

where $\mu_\theta(\mathbf{x}_t, t)$ is the **mean predictor**.

Mean Predictor

How to compute

$$\mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)}[D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]?$$

Q. When $p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_p, \sigma^2 \mathbf{I})$ and $q(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_q, \sigma^2 \mathbf{I})$,

What is $D_{KL}(p \parallel q)$?

[A similar problem to the quiz in the last lecture.]

Mean Predictor

A.

$$p(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_p, \sigma^2 \mathbf{I})$$

$$q(\mathbf{x}) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_q, \sigma^2 \mathbf{I})$$

$$D_{\text{KL}}(p \parallel q) = \frac{1}{2\sigma^2} \|\boldsymbol{\mu}_q - \boldsymbol{\mu}_p\|^2$$

Mean Predictor

How to compute

$$\mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]?$$

$$\mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)} [D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))]$$

$$= \frac{1}{2\tilde{\sigma}_t^2} \mathbb{E}_{t>1, q(\mathbf{x}_t|\mathbf{x}_0)} [\|\mu_\theta(\mathbf{x}_t, t) - \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)\|^2]$$

This weighting term is typically
ignored in practice.

Mean Predictor Training

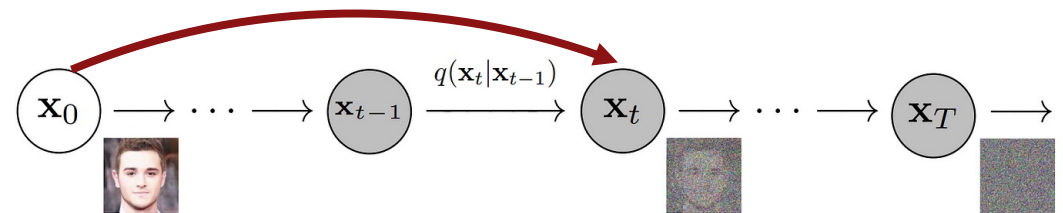
$$\mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), t > 1, q(\mathbf{x}_t | \mathbf{x}_0)} [\|\mu_\theta(\mathbf{x}_t, t) - \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)\|^2]$$

Repeat:

1. Take a random $\mathbf{x}_0$.
2. Sample $t \sim \mathcal{U}(\{1, \dots, T\})$.
3. Sample $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. Compute $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\varepsilon}_t$.
5. Take gradient descent step on $\nabla_\theta \|\mu_\theta(\mathbf{x}_t, t) - \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)\|^2$.

Same as sampling

$$\mathbf{x}_t \sim q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$



Clean-Sample Predictor

$$\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0$$

What if we predict $\mathbf{x}_0$ instead of the mean $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)$?

Let's consider a clean-sample predictor $\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t)$.

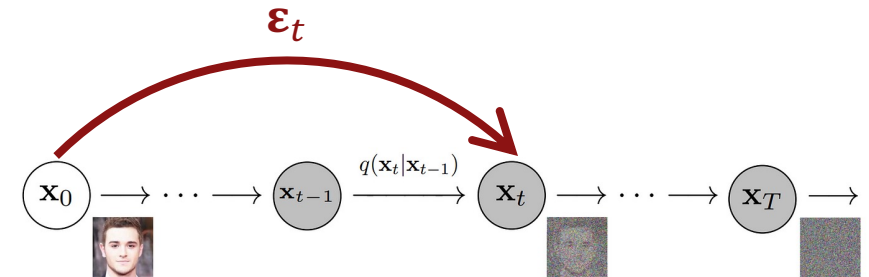
Clean-Sample Predictor

$$\begin{aligned} & \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[D_{KL} \left(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) \right) \right] \\ &= \frac{1}{2\tilde{\sigma}_t^2} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\|\mu_\theta(\mathbf{x}_t, t) - \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)\|^2] \\ &= \frac{1}{2\tilde{\sigma}_t^2} \frac{\bar{\alpha}_{t-1}\beta_t^2}{(1 - \bar{\alpha}_t)^2} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\|\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t) - \mathbf{x}_0\|^2] \\ &= \cancel{\sigma}_t \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\|\hat{\mathbf{x}}_\theta(\mathbf{x}_t, t) - \mathbf{x}_0\|^2] \end{aligned}$$

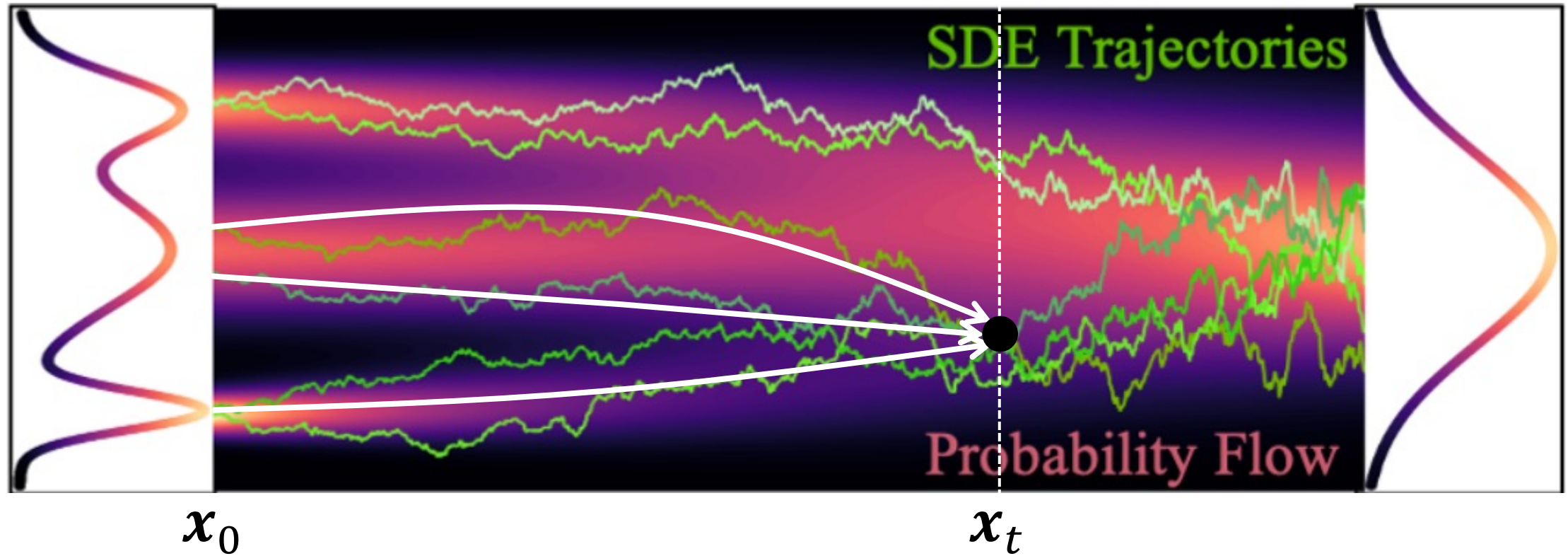
Clean-Sample Predictor

$$\mathbb{E}_{q(x_t|x_0)} [\|\hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t) - \mathbf{x}_0\|^2]$$

- $\mathbf{x}_t$ is sampled from $\mathbf{x}_0$.
- From $\mathbf{x}_t$, predict the *expected* value of $\mathbf{x}_0$ that would result in sampling $\mathbf{x}_t$ from it through the forward jump.



Clean-Sample Predictor



Clean-Sample Predictor

- Note that our goal is to sample $\mathbf{x}_0$ from a standard normal sample $\mathbf{x}_T$ and through latent variables $\mathbf{x}_{T-1}, \mathbf{x}_{T-2}, \dots, \mathbf{x}_1$.
- But for every $\mathbf{x}_t$, we directly predict the expected value of $\mathbf{x}_0$ from $\mathbf{x}_t$.

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$$

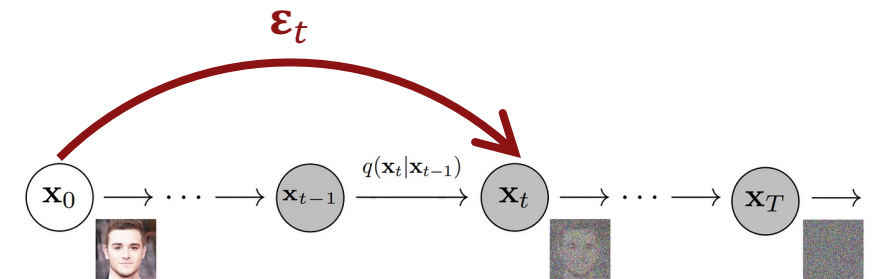
From the forward jump $q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$,

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_t \quad \text{where } \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

If $\mathbf{x}_t$ and $\mathbf{x}_0$ are given, define $\boldsymbol{\epsilon}_t$ as

$$\boldsymbol{\epsilon}_t = \frac{1}{\sqrt{1 - \bar{\alpha}_t}} \mathbf{x}_t - \frac{\sqrt{\bar{\alpha}_t}}{\sqrt{1 - \bar{\alpha}_t}} \mathbf{x}_0$$

Q. Rewrite $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)$ as a function of $\mathbf{x}_t$ and $\boldsymbol{\epsilon}_t$.



$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$$

A.

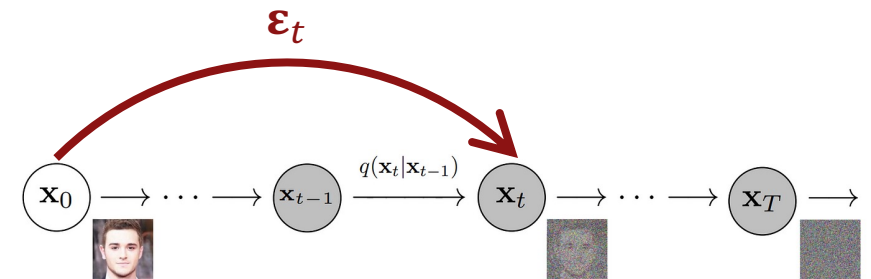
$$\begin{aligned}\tilde{\mu}(\mathbf{x}_t, \boldsymbol{\epsilon}_t) &= \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 \\ &= \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_t \right)\end{aligned}$$

Noise Predictor

$$\tilde{\mu}(\mathbf{x}_t, \boldsymbol{\epsilon}_t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_t \right)$$

What if we predict $\boldsymbol{\epsilon}_t$ instead of the mean $\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)$?

Let's also consider a noise predictor $\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$.



Noise Predictor

$$\begin{aligned} & \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} \left[D_{KL} \left(q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \parallel p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) \right) \right] \\ &= \frac{1}{2\tilde{\sigma}_t^2} \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\|\mu_{\theta}(\mathbf{x}_t, t) - \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0)\|^2] \\ &= \frac{1}{2\tilde{\sigma}_t^2} \frac{(1 - \bar{\alpha}_t)^2}{\bar{\alpha}_t(1 - \bar{\alpha}_t)} \mathbb{E}_{q_{\phi}(\mathbf{x}_t|\mathbf{x}_0)} [\|\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) - \boldsymbol{\epsilon}_t\|^2] \\ &= \cancel{\phi}_t' \mathbb{E}_{q(\mathbf{x}_t|\mathbf{x}_0)} [\|\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) - \boldsymbol{\epsilon}_t\|^2] \end{aligned}$$

Variational Distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$$

Three options to define variational distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$:

1. $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu_{\theta}(\mathbf{x}_t, t), \tilde{\sigma}_t^2 \mathbf{I})$
with the **mean predictor** $\mu_{\theta}(\mathbf{x}_t, t)$.

Variational Distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$$

Three options to define variational distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$:

$$2. \quad p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t} \hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t), \tilde{\sigma}_t^2 \mathbf{I}\right)$$

with the $\mathbf{x}_0$ predictor $\hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t)$.

Variational Distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\sigma}_t^2 \mathbf{I})$$

Three options to define variational distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$:

$$3. \quad p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)\right), \tilde{\sigma}_t^2 \mathbf{I}\right)$$

with the $\boldsymbol{\epsilon}_t$ predictor $\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$.

Variational Distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$

1. Mean predictor $\mu_{\theta}(\mathbf{x}_t, t)$

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mu_{\theta}(\mathbf{x}_t, t), \tilde{\sigma}_t^2 \mathbf{I})$$

2. Clean-sample predictor $\hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t)$

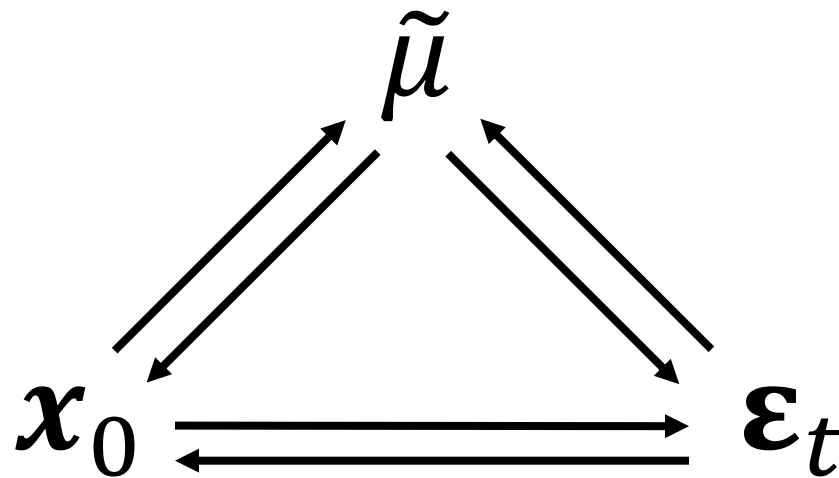
$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1 - \bar{\alpha}_t} \hat{\mathbf{x}}_{\theta}(\mathbf{x}_t, t), \tilde{\sigma}_t^2 \mathbf{I}\right)$$

3. Noise predictor $\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)$

$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)\right), \tilde{\sigma}_t^2 \mathbf{I}\right)$$

Variational Distribution $p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t)$

Given $\mathbf{x}_t$, each of $\tilde{\mu}$, $\mathbf{x}_0$, and $\boldsymbol{\varepsilon}_t$ can be computed from any of the others.



Diffusion Matching Term

In practice, we generally use the ϵ_t predictor $\hat{\epsilon}_\theta(x_t, t)$ since

ϵ_t are standard normal samples

→ well normalized and scaled

→ which makes it easier to train a neural network.

Training

How to train the ϵ_t predictor $\hat{\epsilon}_\theta(\mathbf{x}_t, t)$ with the loss function:

$$\mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), t > 1, q(\mathbf{x}_t | \mathbf{x}_0)} [\|\hat{\epsilon}_\theta(\mathbf{x}_t, t) - \epsilon_t\|^2]$$

Training

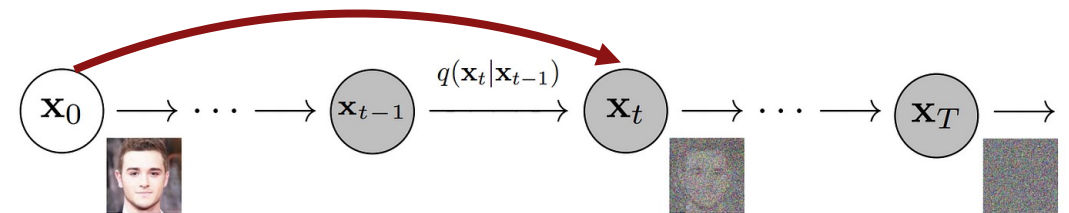
$$\mathbb{E}_{\mathbf{x}_0 \sim q(\mathbf{x}_0), t > 1, q(\mathbf{x}_t | \mathbf{x}_0)} [\|\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) - \boldsymbol{\epsilon}_t\|^2]$$

Repeat:

1. Take a random $\mathbf{x}_0$.
2. Sample $t \sim \mathcal{U}(\{1, \dots, T\})$.
3. Sample $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
4. Compute $\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_t$.
5. Take gradient descent step on $\nabla_{\theta} \|\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) - \boldsymbol{\epsilon}_t\|^2$.

Same as sampling

$$\mathbf{x}_t \sim q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$



Reverse Process (Generation)

How to proceed the reverse process with the **learned** reverse transition distribution:

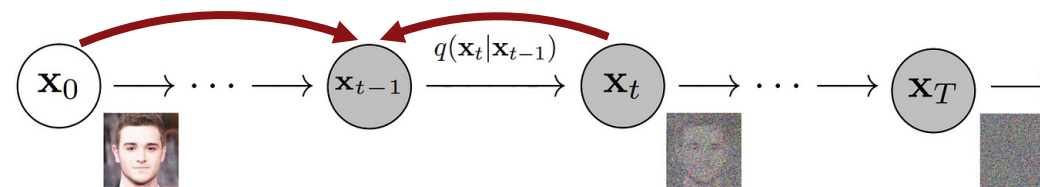
$$p_{\theta}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}}\hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t)\right), \tilde{\sigma}_t^2\mathbf{I}\right)$$

Reverse Process (Generation)

1. Sample $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
2. For $t = T, \dots, 1$, repeat:

1. Compute $\tilde{\boldsymbol{\mu}} = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{x}_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \hat{\boldsymbol{\epsilon}}_{\theta}(\mathbf{x}_t, t) \right)$.
2. Sample $\mathbf{z}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
3. Compute $\mathbf{x}_{t-1} = \tilde{\boldsymbol{\mu}} + \tilde{\sigma}_t \mathbf{z}_t$.

Same as sampling
 $\mathbf{x}_t \sim p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t)$.



Reverse Process (Generation)

At each timestep t , given a noisy sample $\mathbf{x}_t$,

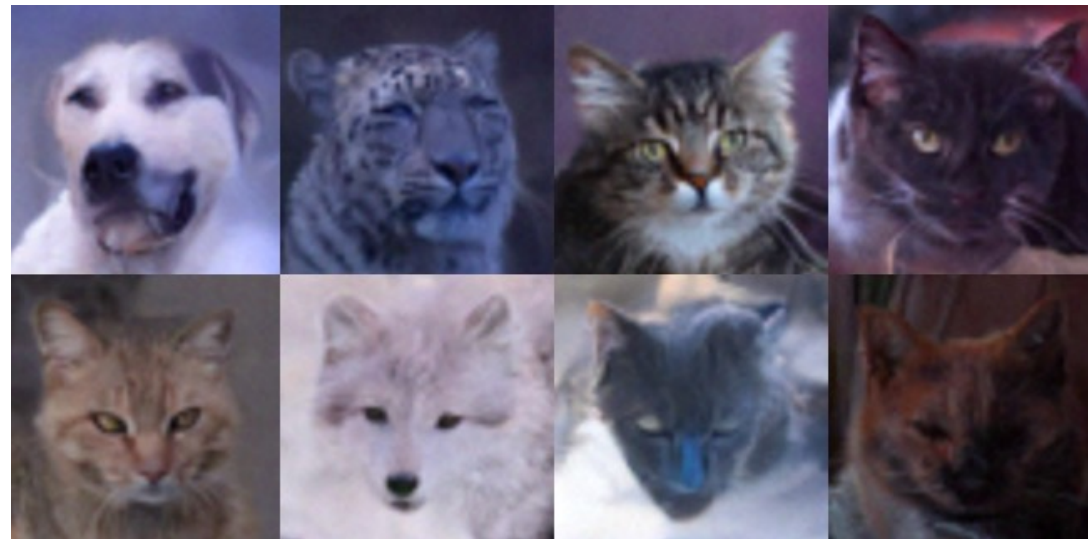
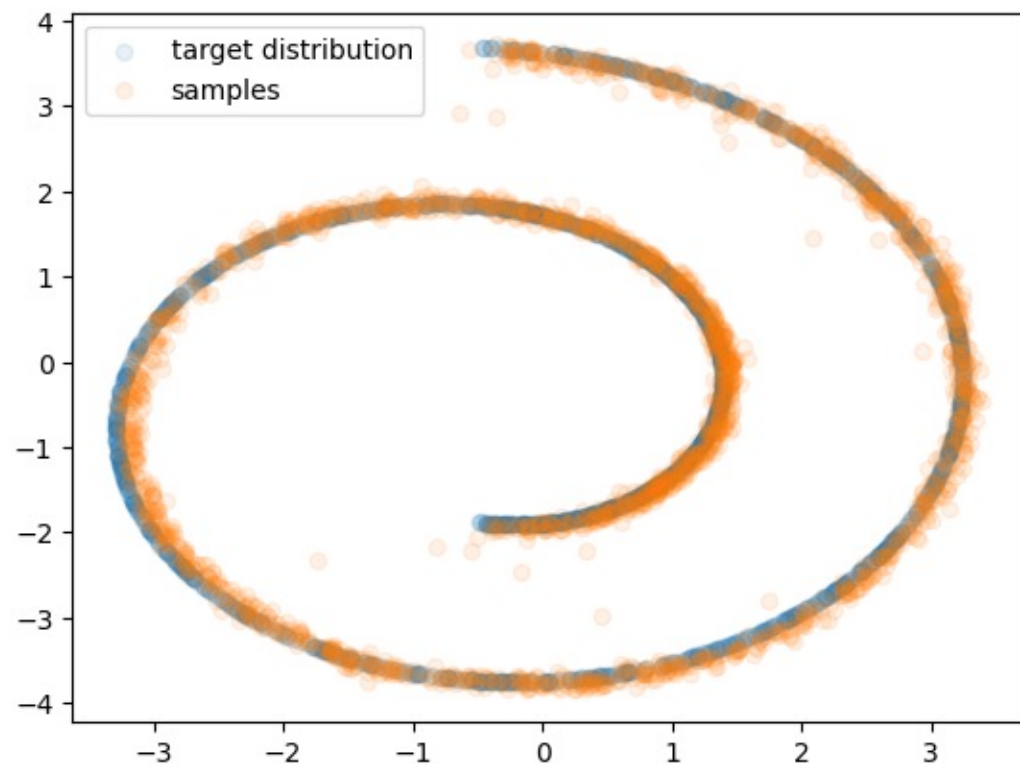
- the network predicts the expected noise $\boldsymbol{\epsilon}_t$.
- From this, the expected clean sample $\mathbf{x}_0$ can be computed from $\mathbf{x}_t$ and $\boldsymbol{\epsilon}_t$.

The denoising (reverse) process can also be viewed as a refinement process!

Reverse Process (Generation)



1st Assignment



Quiz

In the forward process, $q(\mathbf{x}_{t+1}|\mathbf{x}_t)$ is modeled as a Gaussian distribution, and so is $q(\mathbf{x}_T|\mathbf{x}_0)$.

In the reverse process, $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is also modeled as a Gaussian distribution. Does this mean that $p_\theta(\mathbf{x}_0|\mathbf{x}_T)$ is also Gaussian?

Answer

No. In the forward process, the mean of $q(\mathbf{x}_{t+1}|\mathbf{x}_t)$ is a **linear** function of $\mathbf{x}_t$.

In the reverse process, the mean of $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is predicted from $\mathbf{x}_t$ using a **nonlinear** neural network.

Hence, $p_\theta(\mathbf{x}_0|\mathbf{x}_T)$ is not necessarily Gaussian.